Combined Attached Content

Attachment 1

I reviewed the linked repository path and learned its structure and main conclusions.

Repository understood

Repository: `theoneste2060/rwanda_landslide_inventory`
Directory: `rwanda_landslide_ews_v2_deliverables/rwanda_ews_v2`

This is **Version 2 of a Rwanda landslide early-warning-system feasibility and data-readiness assessment**. The main report is dated **October 18, 2026**.

Main files

- `README.md` — setup, interpretation, reproducible commands, and deliverables
- `report.md` / `report.pdf` — full feasibility and evidence report
- `evidence_table.csv` — verified DOI/URL evidence records
- `rema_extraction_table.csv` — page-referenced extraction from the REMA 2022 report
- `inventory_dataset_crosswalk.csv` — comparison of available inventories and datasets
- `data_access_log.csv` — access status, custodians, and contact routes
- `source_findings.md` — source verification notes
- `duplicate_doi_review.csv` — duplicate and alias review
- `input/` — packaged data, dictionaries, QA material, licenses, and source inputs
- `source/` — official REMA report and related source material
- `scripts/` — analysis and reproducibility scripts
- `outputs/` — generated readiness, validation, and audit results

Central scientific conclusion

The project carefully separates three different data products:

1. **The 88-event incident compilation**
    - Event-oriented records

- Useful for source reconciliation and rainfall/event analysis
- Not a complete national landslide census
- Only a subset has precise dates, districts, and linked sites

2. **REMA's 2,544-point shallow-landslide inventory**
   - Used for static landslide susceptibility and historical area-rate analysis
   - Points are mapped initiation locations from Google Earth imagery
   - They are not 2,544 independent, day-dated rainfall-triggering events
   - They cannot directly support a validated rainfall threshold or operational warning probability

3. **The public 7,944-point Kivu-region inventory**
   - Available through Zenodo under CC BY 4.0
   - Potentially useful for regional research
   - Not confirmed to be identical to REMA's 2,544-point dataset
   - Its raw year field requires careful interpretation

What REMA's work provides

REMA's 2022 report provides evidence for:

- Static susceptibility mapping
- Historical landslide affected-area rates
- Exposure overlays using land-use data
- A published logistic-regression susceptibility model
- A reported selected-model AUC of **88.8%**
- Susceptibility-class area rates of approximately:

Susceptibility class	Reported source-area rate
0–10% highest susceptibility	365 m²/year/km²
>10–30%	90 m²/year/km²
>30–60%	18 m²/year/km²
>60–100%	1 m²/year/km²

These are **relative susceptibility and retrospective area-rate measures**, not probabilities that a landslide will occur during a particular rainfall event.

Important limitations

The current material does **not** establish:

- A validated Rwanda rainfall threshold
- Operational alert levels
- Real-data cross-validation of a rainfall-trigger model
- Prospective warning skill
- A confirmed complete national inventory
- The exact REMA 2,544-point layer and full predictor stack
- A transferable future-event AUC of 88–91%

The report also warns that REMA's random 10-fold validation may produce optimistic performance because nearby or related landslides can occur in both training and test folds. Spatial-block and temporal validation are recommended.

Proposed EWS architecture

The project recommends keeping four components separate:

1. **Static susceptibility**
    - Identifies locations where monitoring should be prioritized
2. **Rainfall trigger**
    - Requires locally calibrated rainfall and antecedent-wetness data
3. **Exposure**
    - Identifies people, infrastructure, roads, agriculture, and other assets at risk
4. **Site evidence and response**
    - Includes cracks, movement, sensors, confirmed failures, field checks, and authority actions

The report deliberately does not recommend numerical warning thresholds until rainfall data, event dates, prospective verification, and institutional agreement are available.

Reproducible scripts

The key scripts are:

- `readiness_baseline.py` — evaluates readiness of the 88-event and site datasets
- `build_v2_evidence.py` — rebuilds the evidence table
- `build_access_log.py` — rebuilds data-access records
- `build_dataset_crosswalk.py` — rebuilds inventory comparisons
- `build_rema_extraction.py` — rebuilds REMA page-level extraction
- `inspect_zenodo_layer.py` — audits the public Zenodo shapefile schema
- `replicate_rema_validation.py` — strict real-data susceptibility replication utility
- `sample_size_demo.py` — synthetic rainfall threshold/sample-size demonstration
- `validation_workflow.py` — synthetic validation scaffold

The code intentionally fails closed when the required real REMA data are unavailable. Synthetic outputs are clearly labeled as demonstrations and should not be interpreted as Rwanda model performance.

My working understanding

I will treat this project as a **careful feasibility and evidence-audit package**, not as an already operational landslide warning system.

The most important interpretation is:

> The repository supports a pathway toward a Rwanda landslide EWS, especially static susceptibility and data-readiness work, but it does not yet provide a validated rainfall-trigger warning model or operational thresholds.

I can now use this repository context to help with tasks such as explaining the report, reviewing the Python scripts, interpreting the datasets, improving the EWS design, preparing a proposal, or identifying the next implementation steps. \newpage

Attachment 2

We should **not train a complex machine-learning model using only the 88 records**. The 88 records are too few, and the report indicates that only a smaller subset has strong date/location

quality. The solution is to use the 88 records as a **local calibration and validation dataset**, while obtaining more information from other sources.

First, separate the two models

1. Static susceptibility model

This predicts **where landslides are more likely because of terrain and environmental conditions**.

Potential training data:

- REMA's reported 2,544 shallow-landslide initiation points
- An equal number of background points
- Terrain, geology, land cover, distance-to-river, road distance, curvature, etc.

This is a spatial model. It is not trained from the 88 dated events alone.

However, the exact REMA points and predictor stack must first be obtained and licensed. The public 7,944-point dataset should not automatically be treated as identical to REMA's 2,544 points.

2. Rainfall-trigger model

This predicts **when rainfall may trigger a landslide**.

The 88 records are relevant here, but they are not all equally, example:

- Maximum rainfall over 1 day
- Cumulative rainfall over 3, 7, or antecedent days
- Possibly rainfall intensity over a short duration

A basic model could be:

```
landslide occurrence = rainfall trigger + antecedent rainfall + susceptib
```

Use:

- Penalized logistic regression
- A simple intensity-duration threshold

- Generalized additive models if the dataset becomes larger
- Bayesian logistic regression if uncertainty must be explicit

Avoid deep learning, random forests with many variables, and large neural networks.

The crucial issue: we need both landslide and non-landslide periods

The 88 records are mostly positive reports. A rainfall model needs:

- Times when landslides occurred
- Comparable times when no landslide occurred
- Rainfall measurements or estimates for both

For every confirmed event, we could create a positive event window. Then we need non-event windows from:

- The same districts
- The same rainy seasons
- Similar monitoring periods
- Days with adequate rainfall observation but no recorded landslide

Randomly generating negative rows would be dangerous because “not recorded” does not necessarily mean “no landslide occurred.”

How to use the 88 events

The 88 records should be used in three roles:

1. Local calibration

Use them to determine whether regional or published rainfall thresholds need adjustment for Rwanda.

2. Held-out testing

Do not use every event for training. For example:

- Train on earlier seasons or districts

- Test on later seasons or different districts
- Use leave-one-storm-out validation where possible

3. Uncertainty assessment

The output should initially be expressed as:

- “Low confidence”
- “Exploratory”
- “Needs field confirmation”

Not as a definitive public warning threshold.

Better statistical design for a small dataset

A good early model could be a **Bayesian hierarchical threshold model**:

```
Probability of landslide =
rainfall effect
+ local susceptibility
+ district variation
+ season variation
```

This allows information to be partially shared across districts while still permitting local differences.

For example:

- A national rainfall relationship is estimated
- Each district receives a district-specific adjustment
- The model shrinks unstable district estimates toward the national average

This is more appropriate than fitting a separate model for each district.

Add external records

To improve sample size, we should combine the 88 records with compatible datasets, such as:

- LIWEAR or other Rwanda rainfall-linked inventories

- Published Rwanda landslide threshold studies
- River and meteorological agency records
- District disaster reports
- Road-maintenance and infrastructure failure records
- Satellite-derived landslide detections
- The public Kivu-region inventory, if its dates and definitions are usable

But records must be harmonized first. We should not simply combine datasets with different definitions of:

- Landslide
- Event date
- Location accuracy
- Trigger type
- Inventory completeness
- Observation period

The model should retain a `source` and `data_quality` field so weaker observations do not silently receive the same weight as well-confirmed events.

What not to do

We should not:

- Treat 2,544 mapped points as 2,544 rainfall-triggering events
- Duplicate or synthetically copy the 88 records
- Randomly split highly related records into training and test sets
- Use synthetic data as if it were real training data
- Fit dozens of rainfall and terrain variables to 25–88 events
- Report an operational threshold without prospective testing
- Use random background points as if they were confirmed non-landslide observations

Practical training plan

Stage A — Data preparation

Create one event table containing:

- Event ID
- Date precision
- Location and district
- Source
- Trigger classification
- Rainfall station or gridded-rainfall link
- Susceptibility class
- Data-quality score
- Whether the event is independent from other records

Deduplicate reports referring to the same storm or landslide.

Stage B — Exploratory model

Fit only:

```
1-day rainfall
3-day rainfall
7-day antecedent rainfall
susceptibility class
```

Use strong regularization and wide uncertainty intervals.

Stage C — Validation

Use:

- Leave-storm-out validation
- Leave-season-out validation
- Spatial or district holdout

- Calibration plots
- Sensitivity analysis for uncertain event dates

Stage D — Prospective pilot

Run the model in real time for at least one or more rainy seasons without changing the threshold during evaluation. Record:

- Warnings issued
- Rainfall conditions
- Confirmed landslides
- Missed events
- False alarms
- Data outages

Only after this should thresholds be considered for operational use.

Bottom line

The 88 records are not useless, but they should not be treated as enough for a large predictive model.

The defensible approach is:

> Use the 88 records for local calibration and validation, use the REMA data for static susceptibility, combine compatible external records to enlarge the rainfall-event sample, and begin with a simple, regularized, uncertainty-aware model.

Initially, the system should be a **decision-support prototype**, not an autonomous warning system.